

# Rethinking Image-Scaling Attacks: The Interplay Between Vulnerabilities in Machine Learning Systems

Yue Gao Ilia Shumailov Kassem Fawaz

University of Wisconsin–Madison, Madison, WI, USA

Vector Institute, Toronto, ON, Canada

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## Robustness must hold across the entire vision pipeline

- Vision AI powers phones, cars, healthcare, retail, and safety.
- Deployed models process diverse, real-world images at scale.
- Robustness must hold across full data pipelines, not just models.

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## Every deployed model relies on upstream image scaling

- Deep models require fixed-size inputs (e.g.,  $224 \times 224$ ).
- Upstream scaling maps high-resolution to model size.
- Scaling reweights pixels; kernels vary by algorithm.

## Scaling and model vulnerabilities interact end-to-end

- Two families: adversarial examples and scaling attacks.
- Independent defenses miss joint, end-to-end effects.
- Need to evaluate the entire pipeline together.

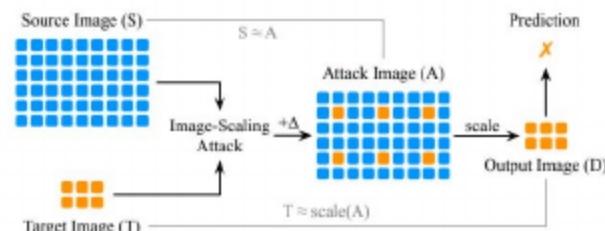


Figure: Illustration: an HR image that looks benign becomes a target image after scaling.

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# Decision-only APIs force query-efficient, gradient-free attacks

- Commercial systems return only labels, no scores.
- No gradients; query budgets are tight.
- Pre-scaling defenses further distort attack signals.

# Attacking the full pipeline uncovers hidden, post-scaling perturbations

- Exploit scaling to hide perturbations post-resize.
- Guide black-box search using scaling-aware noise.
- Bypass pre-scaling defenses with defense-aware tricks.

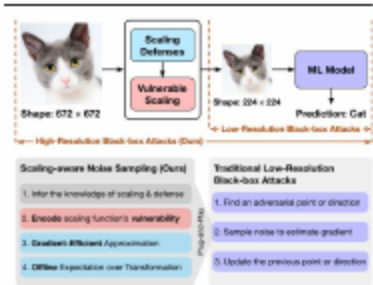


Figure: Scaling-aware noise sampling guides black-box attacks to exploit scaler weaknesses.

# Scaling-aware, defense-aware methods strengthen black-box attacks

- Scaling-aware Noise Sampling (SNS) for decision-based attacks.
- Defense-aware methods: median gradients and offline EOT.
- System-level analysis across models, defenses, and an API.
- Efficient SNS matches precise inversion with lower compute.

# Optimizing in HR for post-scaling effects improves attack efficacy

- Pipeline:  $F = f \circ g \circ h$  (defense, scaling, model).
- Operate in HR space but optimize for post-scaling effect.
- Plug-and-play with HSJ and Sign-OPT attacks.

## Sampling in LR and projecting to HR targets the scaling-hidden subspace

- Sample noise in LR space where the model operates.
- Project LR noise back to HR via scaling pipeline.
- Bias search toward scaling-hidden directions.

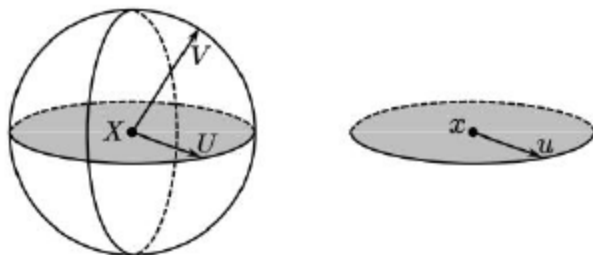


Figure: SNS samples in LR and projects to HR to target scaling-hidden directions.

Navigation icons: back, forward, search, etc.

## Efficient SNS projection guides HSJ and Sign-OPT

- Use projection-matching gradients as an efficient inverse; preserve query budget.
- HR HSJ: SNS informs boundary-direction estimates; HR Sign-OPT: optimize along SNS-guided directions.
- Maintain minimal post-scaling perturbations with no algorithm changes.

Navigation icons: back, forward, search, etc.

## Approximating the median restores usable gradients

- Median flattens response, stalling line search.
- Remedy: trimmed, weighted average approximation.
- Model as masked pooling to target vulnerable pixels.

Navigation icons: back, forward, search, etc.

## Offline EOT captures randomized filtering with zero extra queries

- Direct EOT is query-heavy in black-box.
- Fold randomness into expected scaling offline.
- Zero extra queries; uniform weighting in expectation.

Navigation icons: back, forward, search, etc.

# Joint optimization outperforms sequential pipelines

- Naive: craft LR adversarial, then hide via scaling.
- Joint: optimize end-to-end with SNS guidance.
- Joint approach wins in success and perturbation.

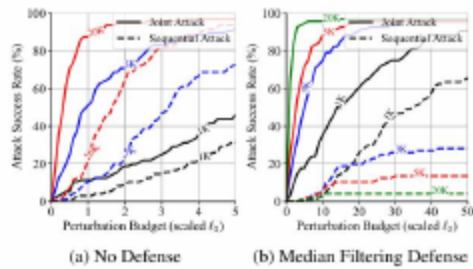


Figure: Joint optimization outperforms a sequential “attack then scale” pipeline.

# ImageNet/CelebA and ResNets provide a realistic evaluation

- Datasets: ImageNet, CelebA.
- Models: ResNet-50/34; inputs resized to 224×224.
- Metrics: scaled  $\ell_2$ , attack success rate (ASR), queries.

# HR variants achieve lower scaled $\ell_2$ and higher ASR without defenses

- Lower scaled  $\ell_2$  at same query budgets.
- Higher ASR for HR HSJ and HR Sign-OPT.
- Ablation: removing SNS erases the gains.

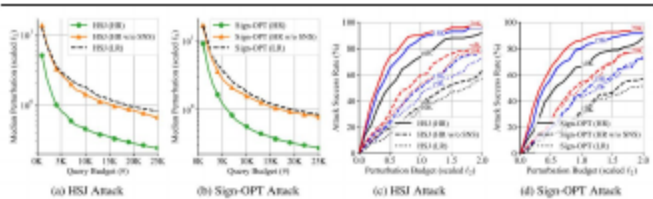


Figure: HR variants achieve lower scaled  $\ell_2$  and higher ASR at the same queries.

# Median filtering does not stop SNS-based HR attacks

- HR retains strong gains under median filtering.
- Improved median gradients restore convergence.
- LR baselines degrade substantially in comparison.

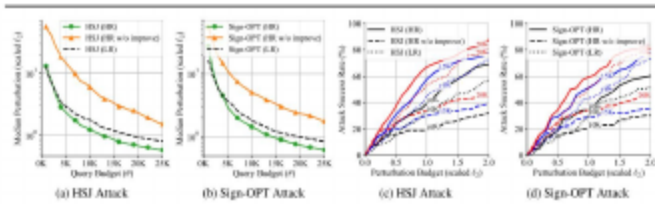


Figure: SNS-based HR attacks retain gains even under median filtering.

# HR HSJ halves $\ell_2$ and roughly doubles ASR on a commercial API

- HR HSJ beats LR HSJ under same query budgets.
- Lower perturbation and higher success rates.
- Scaling vulnerabilities exist in practical systems.

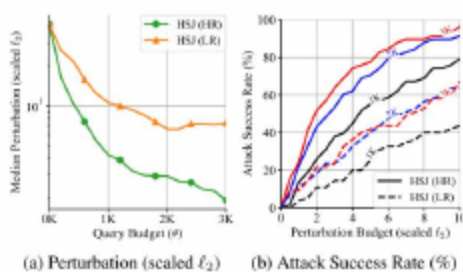


Figure: HR HSJ reduces  $\ell_2$  and doubles ASR on a commercial API.

# HR C&W transfers better to the API without confidence tuning

- HR C&W transfers more effectively to the API.
- Reported scores confirm genuine gains.
- No reliance on inflated confidence settings.

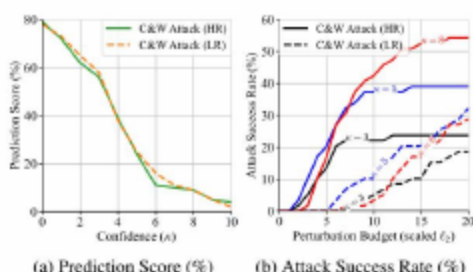


Figure: HR C&W transfers more effectively without tuning confidence.

# Higher HR scaling ratios cut scaled $\ell_2$ and boost ASR across budgets

- Bigger HR inputs yield lower scaled  $\ell_2$ .
- ASR rises sharply with more queries.
- HR dominates LR across budgets and sizes.

| Attacks             | Query = 10K |       | Query = 15K |       | Query = 20K |        |
|---------------------|-------------|-------|-------------|-------|-------------|--------|
|                     | $\ell_2$    | ASR   | $\ell_2$    | ASR   | $\ell_2$    | ASR    |
| HSJ (LR, 224×224)   | 5.26        | 10.8% | 4.16        | 14.0% | 3.58        | 18.3%  |
| HSJ (HR, 672×672)   | 1.73        | 59.6% | 1.37        | 74.5% | 1.17        | 86.2%  |
| HSJ (HR, 1120×1120) | 1.03        | 90.4% | 0.82        | 98.9% | 0.70        | 100.0% |

# On the cloud API, HR HSJ achieves lower $\ell_2$ and higher ASR at equal queries

- HR achieves lower  $\ell_2$  at same query budgets.
- ASR roughly doubles versus LR at 2K–3K queries.
- Confirms scalability to commercial services.

| Attacks  | Query = 1K |       | Query = 2K |       | Query = 3K |       |
|----------|------------|-------|------------|-------|------------|-------|
|          | $\ell_2$   | ASR   | $\ell_2$   | ASR   | $\ell_2$   | ASR   |
| HSJ (LR) | 10.57      | 10.9% | 6.64       | 18.2% | 7.42       | 21.8% |
| HSJ (HR) | 4.24       | 25.9% | 2.93       | 43.1% | 1.92       | 51.7% |

# White-box results: robust-by-design scaling resists, others break

- HR PGD/C&W confirm lower end-to-end perturbations across typical scalers.
- Median filtering can be bypassed with proper gradient approximations.
- Robust-by-design or randomized scaling neutralizes HR advantages.

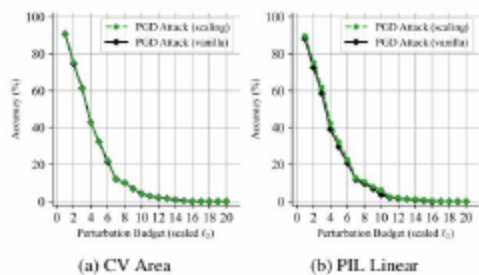


Figure: Robust-by-design scaling removes the exploitable subspace; HR brings no extra gains.

# Robustness is a pipeline property, not a component attribute

- End-to-end HR attacks exploit interactions between scaling and the model.
- Component-only defenses overestimate robustness; SNS generalizes across attacks and datasets.
- Only randomized or robust-by-design scaling is effective in practice.

# Pipeline metrics, broader norms, and detectors remain open

- Pipeline-level robustness metrics and certification.
- Extend SNS to other black-box families, norms.
- Learning-based detectors for subtle, hidden perturbations.

# Questions

- Thank you for your attention!
- Questions and feedback are welcome.

- We thank collaborators, advisors, and supporting institutions.
- We acknowledge open-source libraries and tools used.